Dispatches

Senescence: Detecting an Evolutionary Fingerprint in Plants

The evolution of a senescent decline in the performance of organisms as they grow old is thought to be an unavoidable aspect of life. A genotype–age interaction in performance is the 'fingerprint' of evolved senescence, which has now for the first time been detected in a plant.

Jon E. Brommer

We all are familiar with the senescent decline in performance that comes with age. The existence of this senescent decline seems, at first glance, to refute the power of natural selection: why have individuals performing poorly during old age not been replaced by other, more vigorous ones carrying genes for high performance during this period? Natural selection, Peter Medawar argued [1], did not weed out poor performance at old age simply because it could not. As a cohort of individuals born at the same time ages, their numbers become less and less because of mortality. As a consequence, the force of natural selection acts strongly during early life, when there are still massive numbers of individuals. Selection is, however, but a weak force when it acts on the relatively few ones that do manage to grow old [2,3]. Mutations which reduce performance at old age can hence accumulate in the germ line, and natural selection cannot really do much about this. George C. Williams was quick to realize the implications of Medawar's notion that the forces of natural selection decline during an organism's lifetime. Williams proposed that mutations that increase early-life performance would also be selectively favored when these early-life benefits came at the expense of late-life performance [4]. Hence, mutations coding for 'live fast, die young' performance can invade and replace genes coding for average performance during an organism's lifespan. Williams' hypothesis (formally known as antagonistic pleiotropy) hence views the evolution of senescence as an unavoidable by-product of natural selection strongly favoring early-life performance over late-life performance. In general, the hallmark signature for these evolutionary explanations for senescence is that additive genetic variance (i.e., the variation in breeding values) varies across ages, resulting in a genotype–age interaction or $G\times A$ (Figure 1). Now, a new study by Benoit Pujol and colleagues [5] reported in this issue of *Current Biology* has demonstrated that $G\times A$ occurs in *Silene latifolia* (Figure 1C), thereby demonstrating for the first time that the 'fingerprint' of evolutionary theories of ageing can also be detected in plants.

To date, the main thrust in research on the evolutionary theories of aging is based on laboratory experiments in small insects, involving breeding designs and other techniques to search for signs of the genetic signatures of ageing expected under mutation accumulation versus antagonistic pleiotropy. For example, artificial selection experiments on *Drosophila melanogaster* have convincingly demonstrated a negative genetic correlation between fecundity at early ages and lifespan [6]. This negative genetic correlation is a key element of Williams' hypothesis (Figure 1B), where advantages at early age can only be obtained at the cost of antagonistic declines in late age. Researchers have also recently started to apply quantitative genetic approaches to search for $G\times A$ in wild animals. The existence of senescence in wild animals has long been debated. Lifespan in the wild would simply not be long enough to allow identifying signs of senescence. Nevertheless, abundant demonstrations of age-related declines in reproductive performance traits and survival in wild birds and mammals have amassed, and also — to some extent — for reptiles and insects [7,8]. Thus far, analyses testing for $G\times A$ in wild animals have in the last decade been conducted in four mammalian (including pre-industrial humans) and four avian species [9]. While not all studies find statistically significant evidence, they do demonstrate that it is possible to also detect $G\times A$ in populations where individuals live under natural environmental conditions [10].

It has, nevertheless, been unclear whether evolutionary theories of senescence also apply to organisms other than (wild) animals. Do plants not age? Enviably, a 4,700 year old bristlecone pine *Pinus longaeva* is, in terms of biomarkers of age, like a 20 year old [11]. Can plants withstand senescence and, by extension, demonstrate that the evolutionary theories of senescence are not universally applicable to all organisms? Plants are modular, which allows them to rejuvenate to some extent: fresh shoots carry new leaves and reproductive parts can be formed all over the soma. Another rather fundamental aspect is how to measure the age of a plant. Many plants are clonal, capable of producing new modular parts. Plants can also remain dormant for a long time: do they age during dormancy? The complex life cycle of plants creates scope for selection to not decline with age [3,12], thereby potentially overturning the basic premise underlying the evolution of senescence. Clearly, the study of senescence in plants presents different challenges compared to working with animals. Nevertheless, senescence is not completely alien to plants and has been found to occur in some plant species [13,14].

Quantifying *GxA* as a way to establish whether the evolutionary theories of senescence apply to plants has not been done before now [5]. The long existence of this knowledge gap is highly surprising, since plants are so amenable to quantitative genetic studies. The traditional focus has been on studying survival senescence in plants, rather than exploring the quantitative genetics of ageing [15]. As a consequence, effort has perhaps mainly been poured into the issue of whether senescence exists or not in

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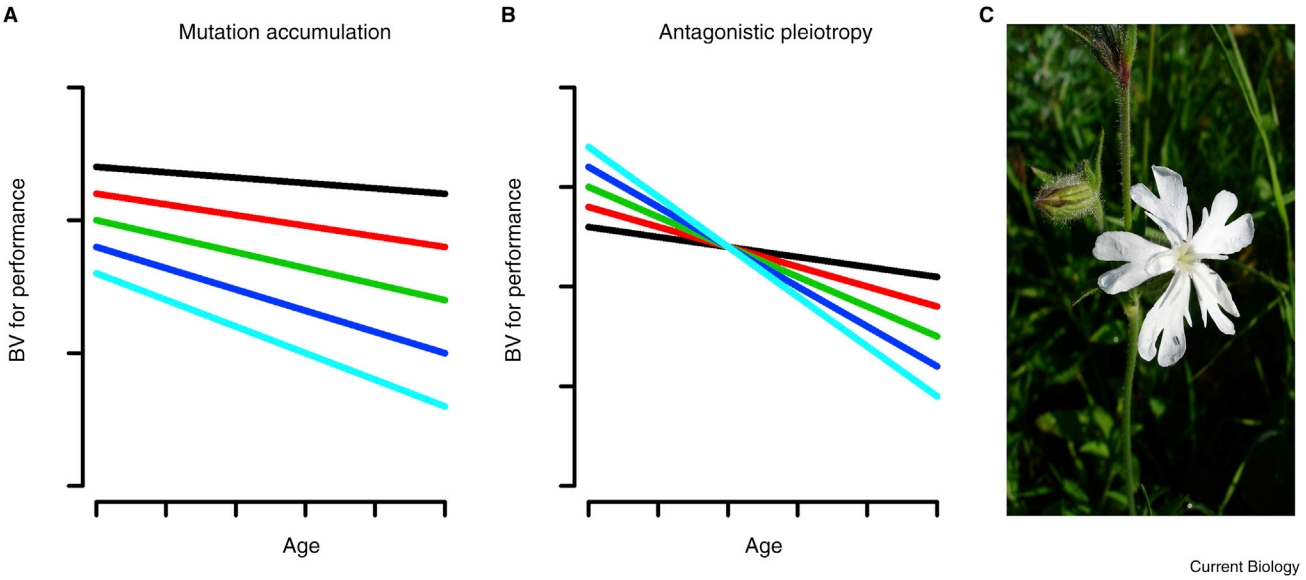


Figure 1. A fingerprint of evolved senescence.

Schematic presentation of the quantitative genetic predictions of evolutionary theories of senescence leading to a decline in breeding value (BV) in a performance trait. (A) A senescent decline is due to the accumulation of mutations deleteriously affecting performance in old age [1]. Over time, different genetic lineages (indicated by their color) have accumulated these deleterious mutations to differing degrees, leading to an increase in additive genetic variance over organismal age. (B) Mutations which increase performance in early age decrease performance in late age (because of pleiotropy) but are able to establish themselves in the population because high early-life performance is selectively favoured. This process inevitably leads to senescence evolving [2,4]. Both theories predict that genetic variance (the 'spread' of the lines) changes over organismal age (genotype–age interaction, $G \times A$), which thereby provides the 'fingerprint' proof for evolved senescence in a population. (C) *Silene latifolia* has now become the first plant where the fingerprint of evolved senescence has been found [5]. (Photograph courtesy of Adrien Favre.)

plants, and whether age-dependent selection indeed would allow the evolution of plant senescence. By focusing on selection, the question of whether the rate of ageing is heritable in plants (i.e., is there *GxA*?) has remained unexplored to date. The design used to demonstrate *GxA* in plant senescence by Pujol and colleagues [5] is not overly complicated. The authors reared plants over three generations by crossing them randomly while keeping track of individual identities, thereby building up a pedigree. They recorded the number and size of flowers each individual produced as it aged. Hence, the focus of this study was on reproductive senescence. The authors then employed evolutionary quantitative genetic approaches commonly used in the study of senescence in pedigreed wild animals [9] to demonstrate *GxA* in reproductive output. That is, there is variation between genetic lineages of *Silene latifolia* in the rate flowering declines over age (Figure 1).

This elegant study hopefully opens up fresh avenues of research into the evolution of senescence in plants, which is a topic of current attention [16]. Clearly, more complex breeding designs could be applied in plants in order to study the role of non-additive genetic variance in plant ageing. Future quantitative genetic studies in plants may also aid in finding information on gerontogene variants underpinning effects of ageing, which has been a successful research program in a wide range of animal taxa [17]. Given the historical focus on survival senescence, a multivariate quantitative genetic perspective on ageing could be interesting. How is the trade-off between components of reproduction and survival resolved during ageing on the genetic level in plants compared to animals? Lastly, plants could be highly amenable to adopting a quantitative genetic framework in a comparative study between populations in order to understand what factors shape differences in age-specific genetic variance components. This aspect has only been explored in some animal populations to date [18]. In general, coming to grips with how environmental conditions influence a decline in performance with age and how the environment modifies the way selection acts on the ageing process are pivotal issues in evolutionary biology [19,20]. I believe that a quantitative genetic perspective on ageing in plants has great potential to further our understanding of whether the evolution of senescence indeed is a universal, unavoidable part of life, as suggested by the classic theories [1,2,4], or whether some organisms can, somehow, escape this ordeal.

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Motor Systems: Reaching Out and Grasping the Molecular Tools

Two recent studies provide important insights into the organization of premotor circuitries, showing that control of highly-specific skilled forelimb movements, such as reaching and grasping, requires activation of specific subpopulations of neurons in the brainstem and spinal cord.

Kuikui Zhou[1], Daniel M. Wolpert[2], and Chris I. De Zeeuw[1,3]

The control of fine finger movements underlying skilled motor behavior has been shown to arise from the development of direct connections from the motor cortex to spinal motor neurons, while more global forelimb tasks are generally considered to depend on the evolutionarily conserved descending pathways mediated by more indirect routes through the brainstem and spinal cord [1,2]. The cerebellum, which is superimposed on these systems, receives internal copies of the motor commands and is required for the precise timing of motor functions, including that of the forelimbs and fingers. To what extent specific forms of forelimb movements are embedded in specific brainstem and spinal cord nuclei has remained elusive because of the technical difficulties of cell and nuclei specific targeting in these regions. Taking advantage of the advent of new viral and optogenetic techniques, two exciting studies from the labs of Silvia Arber [3] and Thomas Jessell [4] provide strong evidence that specific subpopulations of neurons in brainstem and spinal cord of mice are required for voluntary control of reaching and grasping movements.

Where Do We Come From and How to Move Forward?

In the 1960s, Lawrence and Kuypers [5,6] showed that the lateral descending brainstem pathways in both cats and monkeys mediate the capacity for independent use of the extremities, particularly of the monkey hand, while the corticospinal pathways, in addition to controlling the brainstem, allow the fractionation of movements exemplified by the ability to independently control the fingers. In contrast, the ventromedial brainstem pathway forms the basic system by which the brain maintains posture and integrates body–limb movements, such as during locomotion.

Since the discovery of this overall division of the descending brainstem pathways more than half a century ago, many anatomical details have been uncovered (Figure 1), including the identification of neurotransmitters involved [7,8]. The precise topography in these systems has been elusive, as many groups of neurons in the brainstem and spinal cord are difficult to identify by their cyto-architecture, and it has been hard to specifically target them using cell-specific promoters. However, following the revolution in molecular biology, over the last decade new expression patterns of proteins and gene regulating processes have been discovered [9], and, equally important, new technical approaches to exploit these discoveries have been invented. For example, viruses that travel trans-synaptically at single synapses can be transfected into transgenic animals and be turned on and off at will in specific cell groups, marking and/or ablating them by driving expressions of fluorescent and/or toxic probes [2,10]. Moreover, optogenetics can now be used to simultaneously stimulate and/or inhibit multiple specific cell groups with different wavelengths of light, as well as independently control the dendritic tree and remote axon terminals [11].

The Arber and Jessell labs [3,4], which are at the forefront of discovering genes and proteins relevant for the development and function of brainstem and spinal cord, have now exploited this knowledge by applying state-of-the-art viral and optogenetic techniques to advance our understanding of the precise functional topography of the lower motor systems.

Role of MdV in Grasping Types of Movements

Esposito *et al.* [3] demonstrate that the brainstem nucleus medullary reticular formation ventral part (MdV), which is probably part of the lateral system described by Kuypers (Figure 1), specifically targets a subgroup of forelimb-related spinal interneurons and motor neurons that mainly control

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